Where are the Crimes?

STATS 765 - Milestone 2

Aidan Jared

Contents

1 Goal **3**

2 Data Source **3**

3 Data Loading **3**
3.1 Missing/Redacted Data . . . 3
3.2 Date Range . . . 3
3.3 Data Dimensionality . . . 3
3.4 Data Transformations . . . 3

4 Data Exploration **4**
4.1 Distribution of Committed Offenses . . . 4
4.2 Criminal Activity Hotspots . . . 4
4.2.1 By Rate . . . 4
4.2.2 By Total . . . 5
4.3 Census breakdown . . . 5
4.3.1 Home Ownership . . . 5
4.3.2 Location 5 Years ago . . . 6
4.3.3 Education . . . 6
4.3.4 Income . . . 7
4.3.5 Job Industry . . . 7

5 Analytical Plan **7**

6 Appendix **9**
6.1 Setup . . . 9
6.1.1 Load and Format Victimisation . . . 9
6.1.2 Crime Distribution . . . 9
6.1.3 Missing Data . . . 10
6.1.4 Load in Meshblock . . . 10
6.1.5 Load in Census data . . . 10
6.1.6 Combining all Data . . . 10
6.2 Crime Totals . . . 11
6.3 Homeownership . . . 11
6.4 Residency . . . 13
6.4.1 Previous Year . . . 13
6.4.2 Residency Five Years ago . . . 15
6.4.3 Years at Residence . . . 17
6.4.4 Time in NZ . . . 19
6.5 Education . . . 21
6.5.1 Student Type . . . 21
6.5.2 Education Qualifications . . . 23
6.6 Income . . . 26
6.6.1 Source . . . 26
6.6.2 Amount . . . 28
6.7 Job . . . 30
6.7.1 Type . . . 30
6.7.2 Industry . . . 32
6.7.3 Occupation . . . 34
6.7.4 Sector . . . 36

1 Goal

The main goal of this project is to see what features in census Statistical Area 2 (areas of 1,000 to 4,000 people) are the highest predictors of crime. This EDA is focusing on part two of the 2023 Census totals by topic for individuals by SA2 dataset.

2 Data Source

The data for this analysis is coming from 3 main sources:

Dataset	Detail	Dimensions
Census	2023 Census totals by topic for individuals by SA2 Part 2	2395×862
Meshblock	Mapping table for meshblock to SA2 lookup	57539×55
Victimisations	Victimisation data with meshblock and timestamp	1133581×17

The ANZSOC_Full_data is from the NZ police, which has the place and month the reported victimization took place, and a meshblock to the SA2 table to join the police data. data with the census data, and for this write-up, I am looking at the main means of travel to work and the population change by ethnic group from the 2023 New Zealand census.

3 Data Loading

3.1 Missing/Redacted Data

Using the documentation from StatsNZ, any -999 is listed as a redacted value due to confidentiality, and -997 for no response. Since we do not have extra information to input these values, we can treat them as `NA`. In the Victimisation data, there are 7497 meshblocks and area units that are redacted, so we cannot use those as well. There are also a few SA2s that have 0 recording for the census; once again, this can be ignored since there is no way of imputing any possible data to fill these empty spots, and these are likely areas with no residential population and are outliers.

3.2 Date Range

The Victimisation data covers the range of 8/2021 to 7/2025, while the census has values for 2018 and 2023. Therefore, to get the most accurate predictions from the data, we are only going to look at data from 2023. The Victimisation data requires us to drop all columns that did not occur in 2023. Given this reduction in data size and increasing the likelihood of some Regions having none to a few data points, we also decided to focus on the 4 major cities of Auckland. This brought the victimization data to 173203 observations. We have also applied this date range to the census data and dropped all columns that had data for 2013 and 2018.

3.3 Data Dimensionality

The census data has over 800 features, but a lot of these features are from the 2013 and 2018 census; we can drop those, as well as any features describing the land area. That brings us down to 304 features that are broken down into multiple sections, such as individual home ownership, years at residence, study participation, and income. Each of these sections has a total number of responses and value, indicating the number of people who fall into each category. Removing the missing data from the victimization data and filtering down to the relevant columns brings us to 7 features and 173203 rows. While the Meshblock we just keep the 2 relant columns but keep all the rows.

3.4 Data Transformations

In order to best reflect how crime is influenced by the neighborhood features, the three datasets were joined together to make a single table. In order not to have a few individual locations dominate the graphs, it was decided to look

at the victimization rate instead of the totals, so it was divided by the total population of each SA2 to find the rate for that area.

Dataset	Original Size	Final Size
Victimisation	1133581×17	173203×7
Meshblock	57539×55	57539×2
Census Totals by Topic	2395×862	2395×304
Main EDA Data		3802×307

4 Data Exploration

4.1 Distribution of Committed Offenses

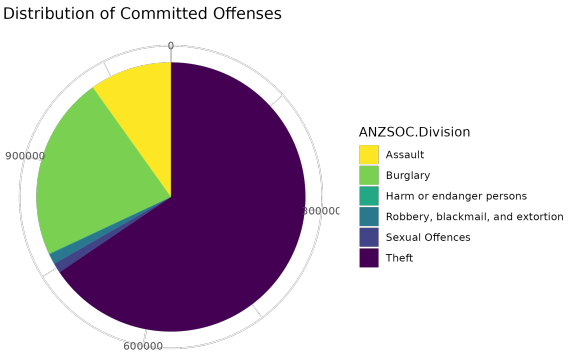


Figure 1: Distribution of Committed Offenses

The majority of crimes are Theft by a large margin, while Sexual Offences and Robbery and blackmail are the least common types of crime. This breakdown is understandable since theft is a less serious type of crime that could just happen with items left out in public areas or people being pickpocketed in crowded areas.

4.2 Criminal Activity Hotspots

4.2.1 By Rate

Statistical Area 2 Name	Total Population	Total Victimisation	Crime Rate
Botany Central	30	1016	3387%
Mount Wellington Industrial	36	267	742%
Addington North	96	379	395%
Riccarton Central	270	853	316%
Sylvia Park	585	1618	277%

Most of the areas with high rates seem to be areas with a low residential population but a high amount of traffic, such as parks and business centers.

4.2.2 By Total

Statistical Area 2 Name	Total Population	Total Victimisation	Crime Rate
Te Rapa South	852	1909	224%
Hamilton Central	3582	1777	49.6%
Sylvia Park	585	1618	277%
Linwood East	10800	1085	10%
Botany Central Park	30	1016	3387%

The areas with high totals of victimizations look to have some overlap with the areas of high rates, but there are now a few more highly populous areas in the mix.

4.3 Census breakdown

Due to the large number of columns and census groups, this only highlights a few of the most interesting trends.

4.3.1 Home Ownership

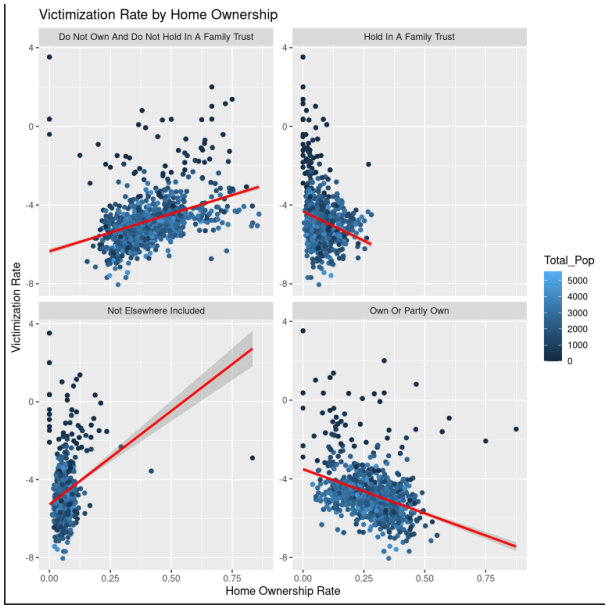


Figure 2: Log Crime Rate by Homeownership

Homeownership shows 3 major trends in victimization, as the rate of not owning in an SA2 increases, so does the victimization rate. This is the inverse for homes that are held in a family trust or owned, which show a downward trend. These trends are most likely correlated with the household wealth, such that areas with more ownership tend to have higher household wealth. There might also be a correlation with geographic location, with more people renting in city centers, while more people own their houses if they are not in the city center. These trends indicate that these features are worth including in the future model.

4.3.2 Location 5 Years ago

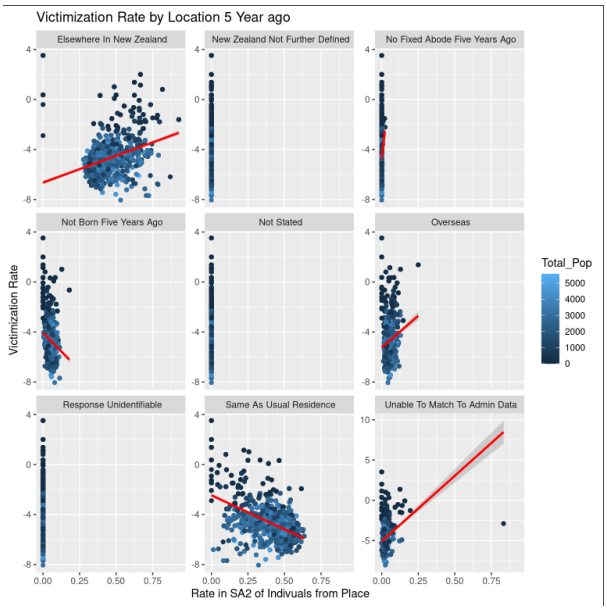


Figure 3: Log Crime Rate by Location 5 Years Ago

A clear trend is that if individuals were located in the same SA2 as they were five years ago, there is a downward trend in victimization. This might indicate that people feel safer in neighborhoods with less crime, so they are more likely to stay. There is an upward trend for people who were located elsewhere in New Zealand five years ago. Since this EDA is looking at the 4 major cities, this trend might be related to individuals recently moving into the major cities (such as students or recent graduates) and being located in more centralized areas. Since there is such a distinct trend long-term occupancy is worth including in the model

4.3.3 Education

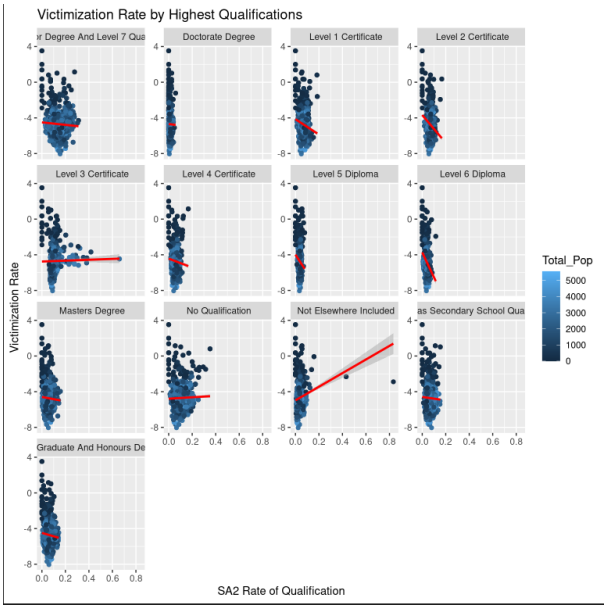


Figure 4: Log Crime Rate by Education Level Rate

Education only shows slight trends in the data with minor downward trends for having and education level. However, the data is so clustered due to the spread of possible values that it cannot be directly determined using this graph if

there is an overall trend.

After grouping some categories together into larger sections, the trends are slightly more pronounced but still very slight. The main trend the graph shows is that SA2 with higher rates of any form of education qualification reduces the rate of Victimisation. But since this trend is so slight, this category can be ignored.

see in Appendix

4.3.4 Income

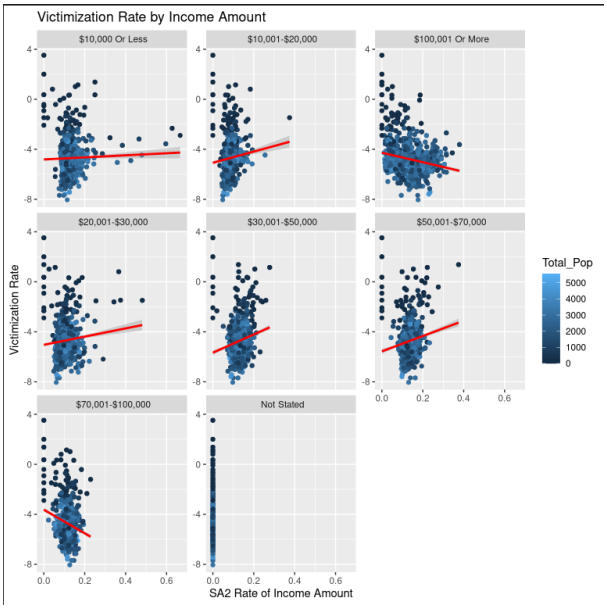


Figure 5: Log Crime Rate by Income Rate

Outside of making $\$70,001+$ per year, there seems to be an upward trend in the victimization rate for all income brackets. While it makes sense that more wealthy SA2 would have a low crime rate, what is interesting is that the general trend seems to be an increase in the crime rate when there are more people with incomes between $\$10000 - \$70000$. Since the highest crime type is theft, this might indicate that in the SA2 with lower income rates, there might be less theft, which then increases as the area's income increases up to the super-rich, who probably have more security and are not targeted as much. With this divide, it might be worth combining the data in these columns down to lower, middle, and high income and using those in the model, but these features are not worth looking at independently.

4.3.5 Job Industry

Most of the job types have a slight downward trend or no discernible trend; the main exception is Accommodation and Food Service. This may be due to such roles having lower salaries, which has the individuals living in less affluent areas and/or being frequently exposed to the public. By limiting this analysis to just the major cities, there may be trends not being captured, such as areas with much higher mining employment and/or gas extraction. While this is an interesting result, it does not work when adding to the model, since in most cases the rate of occupation type has no major influence on the crime rate.

5 Analytical Plan

The next step would be establishing relationships between all predictors against a normalized count of victimizations (/ 1000 Households). There is a strong potential for income and crowding to explain the number of victimizations, hence the possibility (polynomial) of a non-linear relationship would be explored. All the socio-economic factors, such as vehicle ownership and access to telecommunication, should be examined in correlation with income and crowding. Establish a geographic context by comparing the patterns in the city center, suburban, and rural settings.



Figure 6: Log Crime Rate by Industry Rate

Observe the outcome of the backward model selection based on these factors. Build a regression model to predict potential high-risk areas based on predictors. Since most explanatory variables here are numeric, XGBoost may be used. Observe the effects of excluding outliers. Identify and implement ways for effective visual geographical reporting. Use LASSO and RIDGE regression to find the best combination of variables and make the model more general.

6 Appendix

6.1 Setup

```
library(Hmisc, include.only = c('describe'))
library("tidyverse")
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(grid)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
library(hexbin)
```

6.1.1 Load and Format Victimisation

```
# Read in crime data and filter to relevant columns
crime_data = read.csv("../data/ANZSOC_Full Data_data.csv") |>
 mutate(Year.Month = my(Year.Month)) |>
 mutate(Month = month(Year.Month), Year = year(Year.Month))|>
 select(ANZSOC.Division, Victimisations, Meshblock,
 Territorial.Authority, Month, Year, Year.Month)
# describe(crime_data)

focused_crime = crime_data |> # limmiting data to the 4 major NZ cities
  filter(Year == 2023 & Territorial.Authority
  %in% c("Auckland.", "Wellington City.", "Hamilton City.", "Christchurch City."))
# describe(focused_crime)
```

6.1.2 Crime Distribution

```
# Pie chart of crime type distribution
crime_data |>
 group_by(ANZSOC.Division) |>
 dplyr::summarize(count_by_subdivision = sum(Victimisations,
    na.rm = T),
    .groups = "drop") |>
 ggplot(aes(x = "",
               y = count_by_subdivision,
               fill = `ANZSOC.Division`)) +
 geom_bar(stat = "identity", width = 1) +
```

```
 scale_fill_viridis_d(direction = -1) +
 coord_polar("y", start = 0) +
 labs(
        title = "Distribution of Committed Offenses",
        x = "",
        y = ""
    ) +
 theme_minimal()
```

6.1.3 Missing Data

```
# number of missing locations
missing_loc = crime_data |> filter(Meshblock < 0) |> nrow()
cat("missing location data: ", missing_loc,'\n')
```

```
## missing location data:  7497
```

6.1.4 Load in Meshblock

```
# load in meshblock data
names = c("MB2023_V1_00" = "Meshblock", "SA22023_V1_00" = "SA2")
meshblock_data = read.csv(
  "../data/meshblock-higher-geographies-2023-generalised.csv"
) |>
 select(MB2023_V1_00, SA22023_V1_00) |>
 rename(
         SA2 = SA22023_V1_00,
         Meshblock = MB2023_V1_00)

# describe(meshblock_data)
```

6.1.5 Load in Census data

```
# load in census
census = read_csv(
  "../data/2023_Census_totals_by_topic_for_individuals_by_SA2_299028350473552691.csv",
  na = c("", "NA", -999, -997), show_col_types = FALSE) |>
 select(!contains("2018")) |> # remove unneccary dates
 select(!contains("2013")) |>
 select(!contains("Shape")) |>
 select(!contains("kilometres")) |>
 rename(SA2 = `Statistical area 2 (SA2) 2023 code`) |>
 rename(Total_Pop = `Subject pop: Census usually resident population count, Year: 2023, Measure: Count, Va

ncol(census)
```

```
## [1] 304
```

6.1.6 Combining all Data

```
# joining crime to SA2
crime_SA2 = inner_join(
 focused_crime,
 meshblock_data,
                      by = "Meshblock")
```

```
total_crime_sa2 = crime_SA2 |>
 group_by(SA2, Territorial.Authority, ANZSOC.Division) |>
 summarise_at("Victimisations", sum) |>
 inner_join(census, by = "SA2")
```

6.2 Crime Totals

```
totals = total_crime_sa2 |>
 group_by(`Statistical area 2 (SA2) 2023 name`) |>
  filter(Total_Pop > 1) |>
 summarise(total_pop = sum(Total_Pop, na.rm = TRUE),
           total_vit = sum(Victimisations, na.rm = TRUE),
           crime_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE) * 100)
totals |> arrange(desc(crime_rate))
```

```
## # A tibble: 928 x 4
##    `Statistical area 2 (SA2) 2023 name` total_pop total_vit crime_rate
##    <chr>                                    <dbl>     <int>      <dbl>
##  1 Botany Central                              30      1016      3387.
##  2 Mount Wellington Industrial                 36       267       742.
##  3 Addington North                             96       379       395.
##  4 Riccarton Central                          270       853       316.
##  5 Sylvia Park                                585      1618       277.
##  6 Te Rapa South                              852      1909       224.
##  7 Inlet Waitemata Harbour                      9        13       144.
##  8 Rosebank Peninsula                          27        39       144.
##  9 Wesley South                               144       206       143.
## 10 Rongotai                                   192       270       141.
## # i 918 more rows
```

```
totals |> arrange(desc(total_vit))
```

```
## # A tibble: 928 x 4
##    `Statistical area 2 (SA2) 2023 name` total_pop total_vit crime_rate
##    <chr>                                    <dbl>     <int>      <dbl>
##  1 Te Rapa South                              852      1909     224.
##  2 Hamilton Central                          3582      1777      49.6
##  3 Sylvia Park                                585      1618     277.
##  4 Linwood East                             10800      1085      10.0
##  5 Botany Central                              30      1016    3387.
##  6 St Lukes                                  9885       981       9.92
##  7 Kirikiriroa                               4110       929      22.6
##  8 Quay Street-Customs Street                9810       915       9.33
##  9 Sydenham Central                          1305       861      66.0
## 10 Riccarton Central                          270       853     316.
## # i 918 more rows
```

6.3 Homeownership

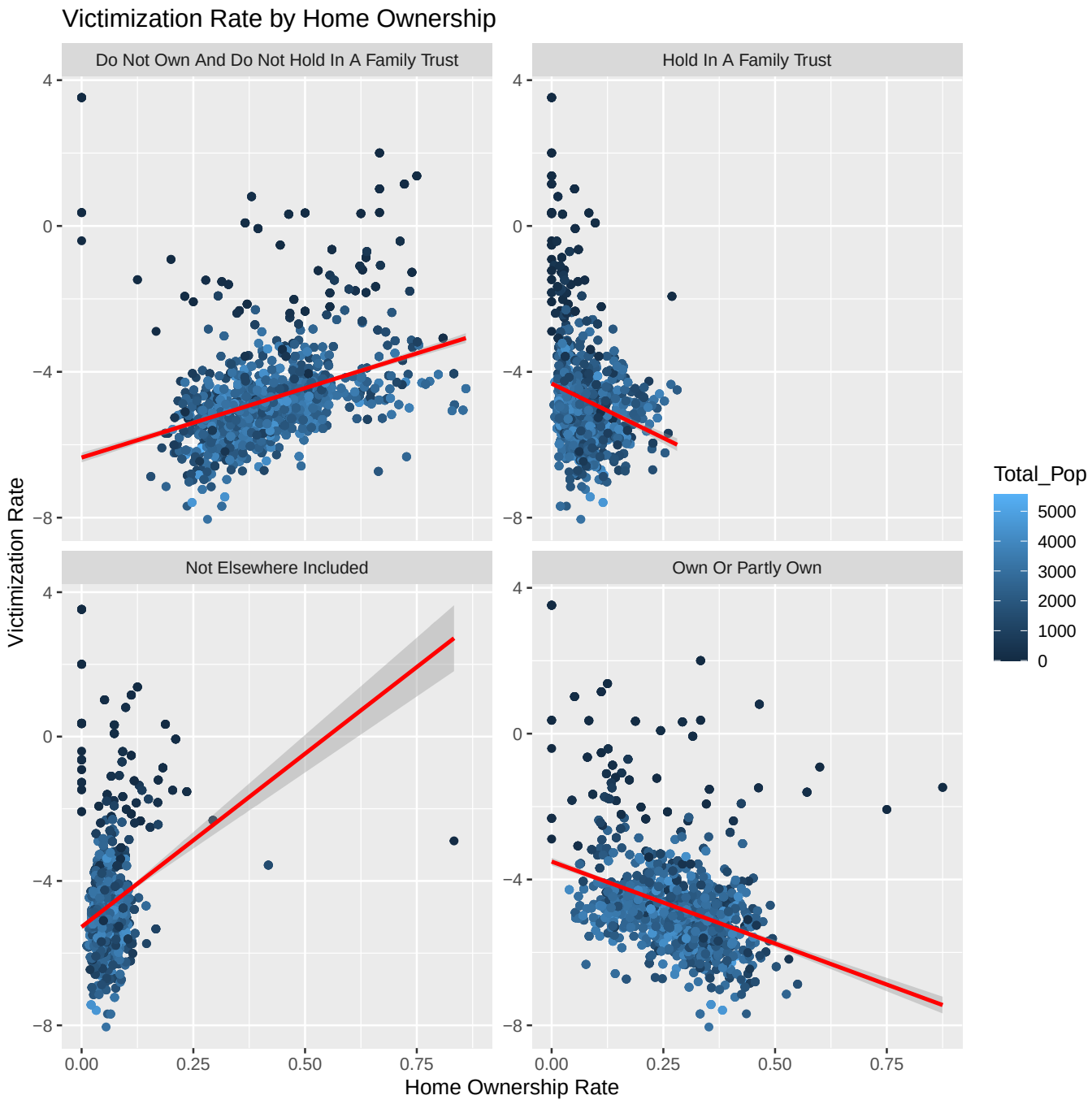
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Census usually resident population count aged 15 years and over, Year: 202
                "pop over 15 \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
```

```
 mutate(
 across(starts_with("pop_over_15_"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
 pivot_longer(
    cols = starts_with("pop_over_15_"),
    names_to = "home_ownership_type",
    values_to = "proportion"
  ) |>
 mutate(
    home_ownership_type = str_remove(home_ownership_type, "pop_over_15_"),
    home_ownership_type = str_replace_all(home_ownership_type, "_", " "),
    home_ownership_type = str_to_title(home_ownership_type)
  ) |>
 select(c("home_ownership_type", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(home_ownership_type, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ home_ownership_type, scales = "free_y") +
 labs(
    x = "Home Ownership Rate",
    y = "Victimization Rate",
    title = "Victimization Rate by Home Ownership"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.4 Residency

6.4.1 Previous Year

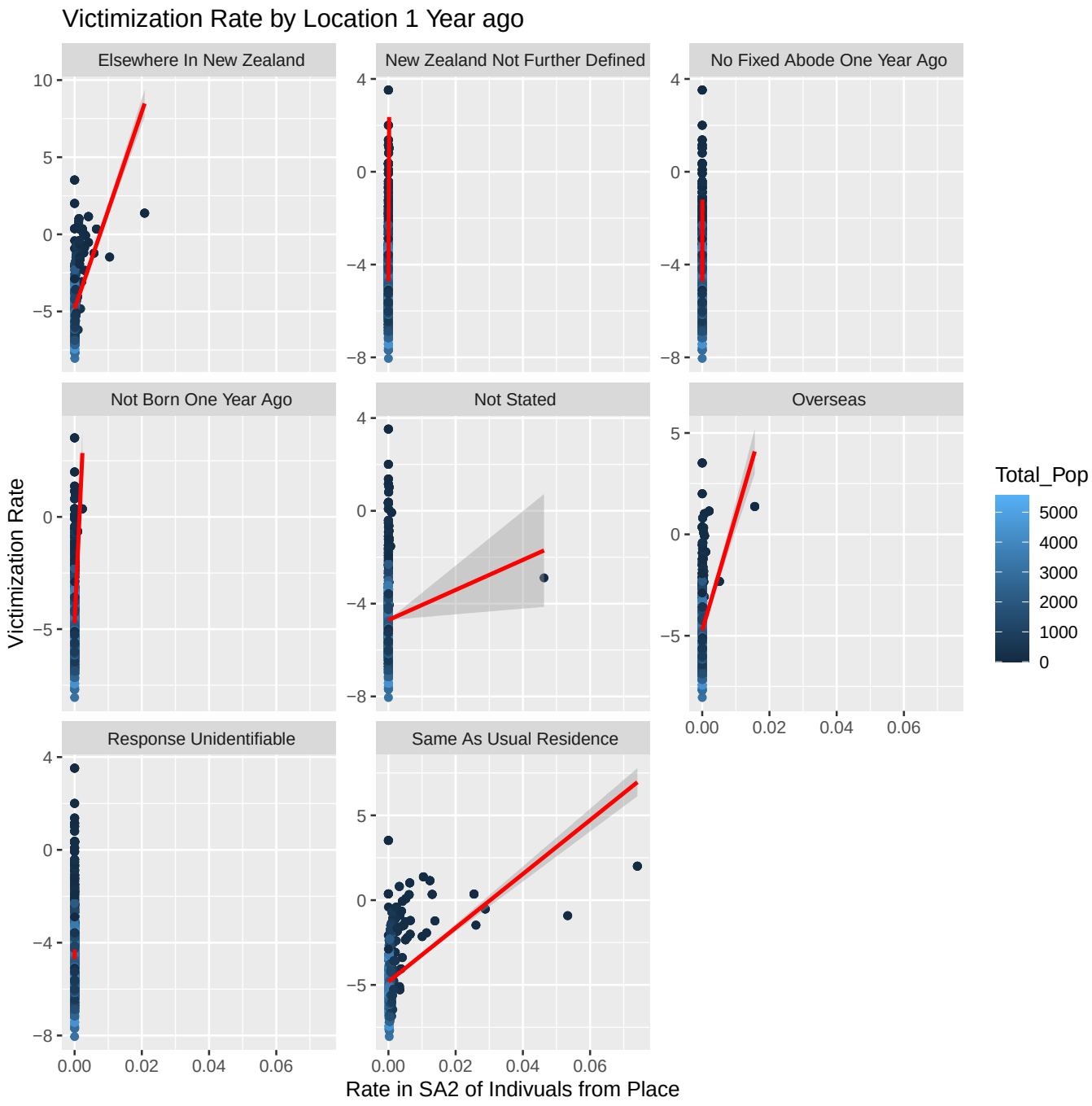
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Census usually resident population count, Year: 2023, Measure: Count, Var1
                "usual_residence_1_yr_ago \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("usual_residence_1_yr_ago"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
```

```
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
 mutate(
 across(starts_with("usual_residence_1_yr_ago"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE))
  ) |>
 pivot_longer(
    cols = starts_with("usual_residence_1_yr_ago"),
    names_to = "comes_from",
    values_to = "proportion"
  ) |>
 mutate(
    comes_from = str_remove(comes_from, "usual_residence_1_yr_ago"),
    comes_from = str_replace_all(comes_from, "_", " "),
    comes_from = str_to_title(comes_from)
  ) |>
 select(c("comes_from", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(comes_from, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ comes_from, scales = "free_y") +
 labs(
    x = "Rate in SA2 of Indivuals from Place",
    y = "Victimization Rate",
    title = "Victimization Rate by Location 1 Year ago"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.4.2 Residency Five Years ago

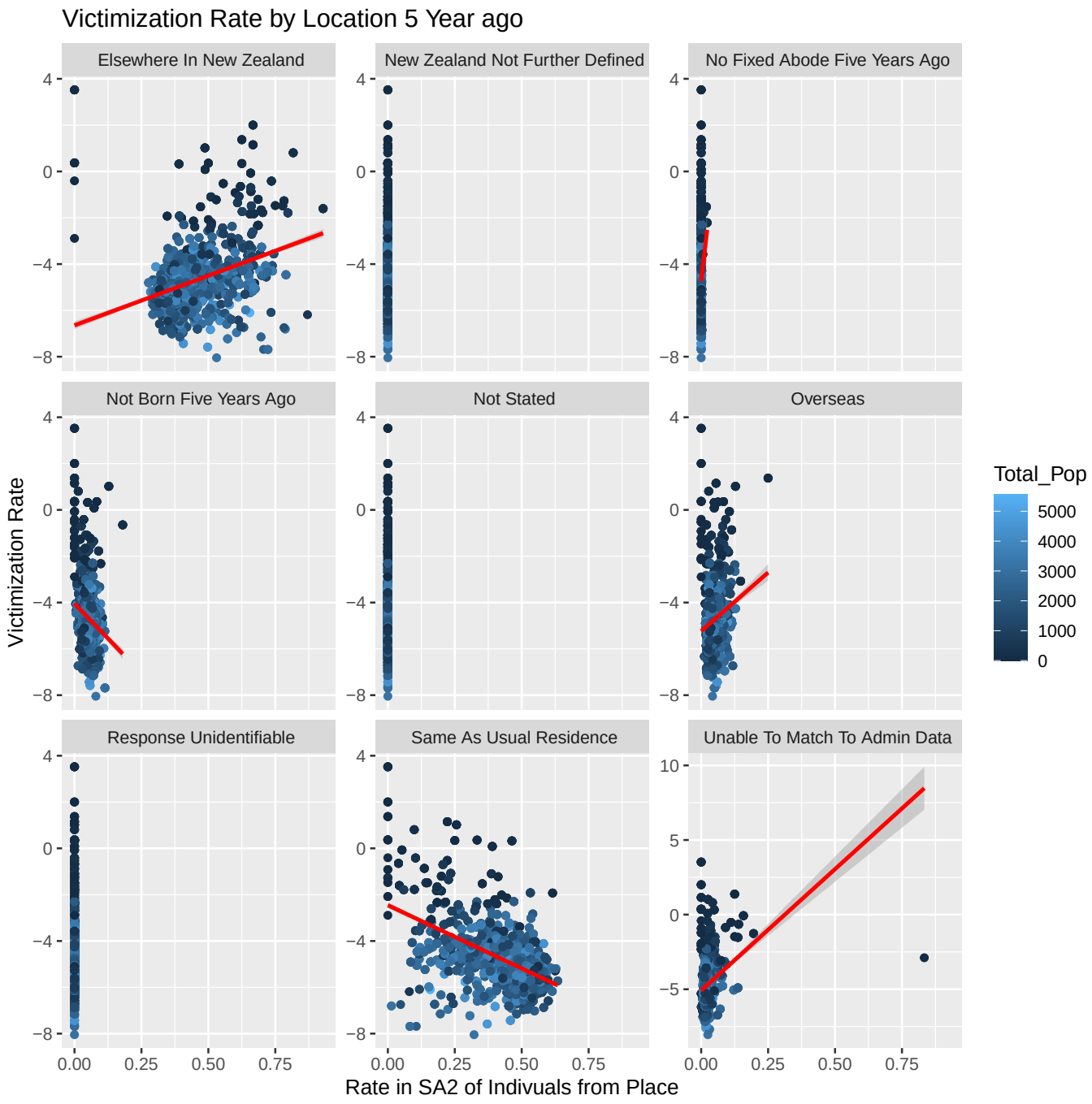
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Census usually resident population count, Year: 2023, Measure: Count, Var1
                "usual_residence_5_yr_ago \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("usual_residence_5_yr_ago"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
```

```
 pivot_longer(
    cols = starts_with("usual_residence_5_yr_ago"),
    names_to = "comes_from",
    values_to = "proportion"
  ) |>
 mutate(
    comes_from = str_remove(comes_from, "usual_residence_5_yr_ago"),
    comes_from = str_replace_all(comes_from, "_", " "),
    comes_from = str_to_title(comes_from)
  ) |>
 select(c("comes_from", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(comes_from, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ comes_from, scales = "free_y") +
 labs(
    x = "Rate in SA2 of Indivuals from Place",
    y = "Victimization Rate",
    title = "Victimization Rate by Location 5 Year ago"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.4.3 Years at Residence

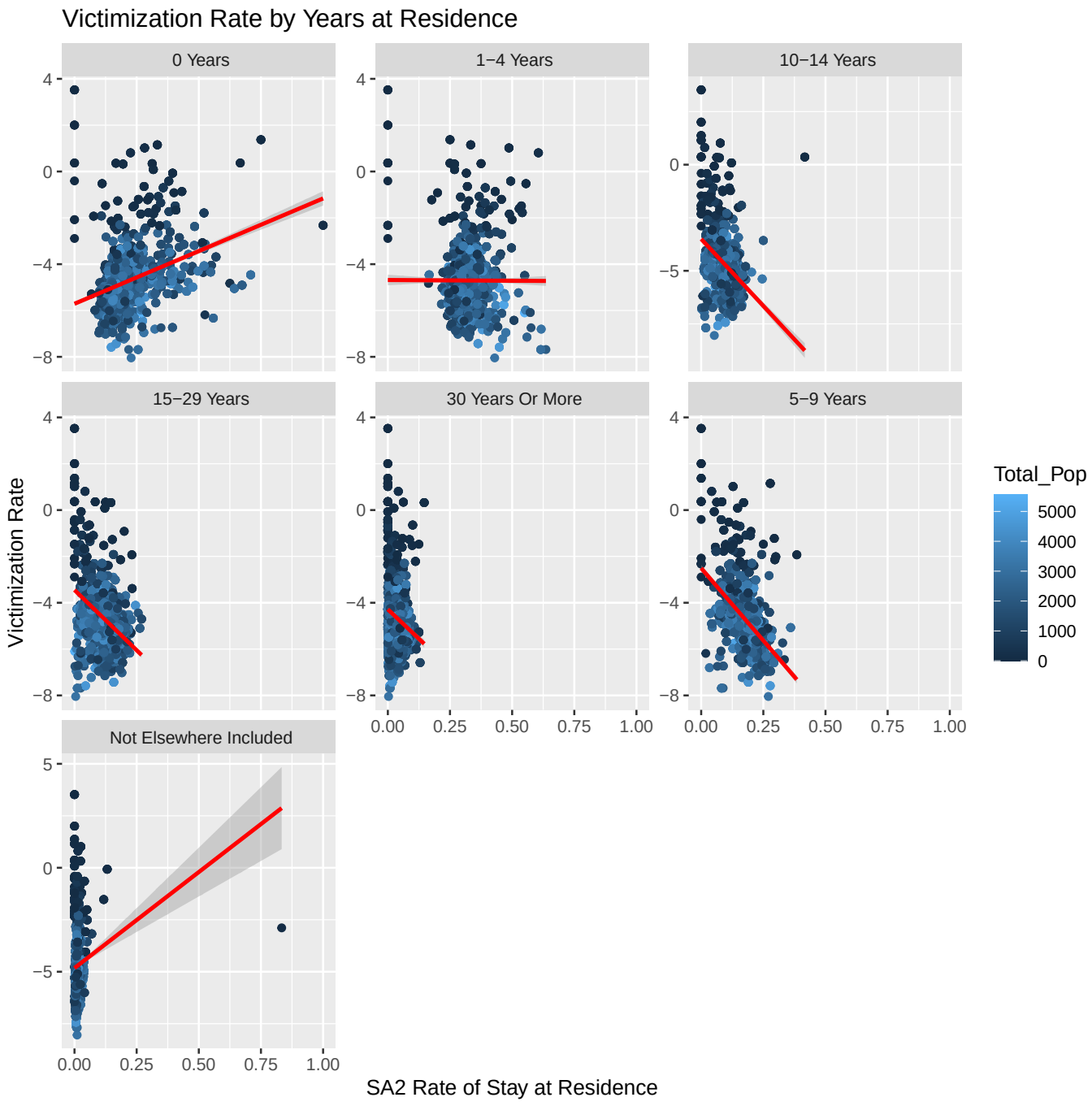
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Census usually resident population count, Year: 2023, Measure: Count, Var1
                "years_at_residence \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("years_at_residence"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
```

```
 pivot_longer(
    cols = starts_with("years_at_residence"),
    names_to = "years_at_residence",
    values_to = "proportion"
  ) |>
 mutate(
    years_at_residence = str_remove(years_at_residence, "years_at_residence"),
    years_at_residence = str_replace_all(years_at_residence, "_", " "),
    years_at_residence = str_to_title(years_at_residence)
  ) |>
 select(c("years_at_residence", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(years_at_residence, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ years_at_residence, scales = "free_y") +
 labs(
    x = "SA2 Rate of Stay at Residence",
    y = "Victimization Rate",
    title = "Victimization Rate by Years at Residence"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.4.4 Time in NZ

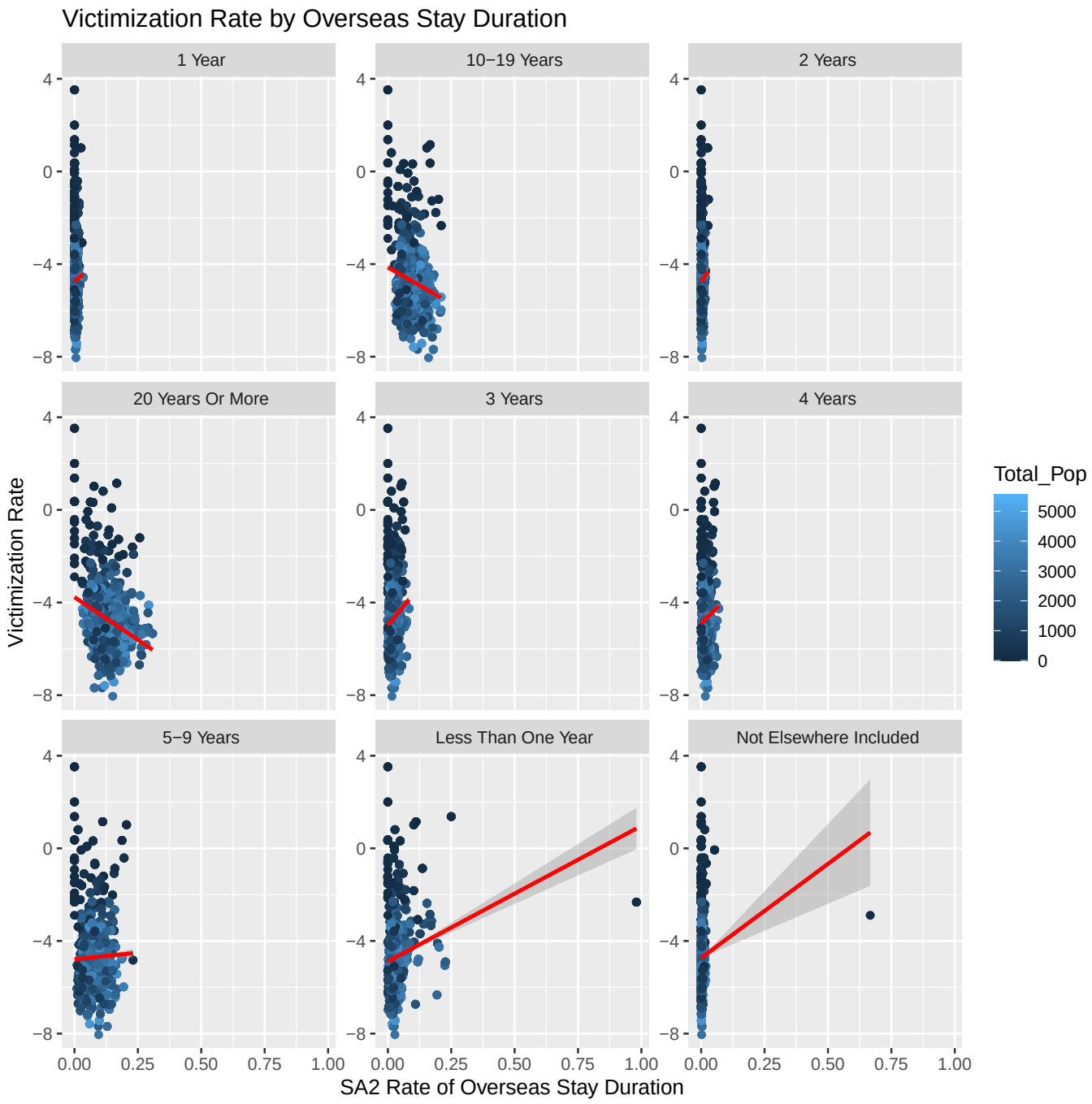
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Overseas-born census usually resident population count, Year: 2023, Measur
                "overseas_stay_duration \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("overseas_stay_duration"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
```

```
 pivot_longer(
    cols = starts_with("overseas_stay_duration"),
    names_to = "overseas_stay_duration",
    values_to = "proportion"
  ) |>
 mutate(
    overseas_stay_duration = str_remove(overseas_stay_duration, "overseas_stay_duration"),
    overseas_stay_duration = str_replace_all(overseas_stay_duration, "_", " "),
    overseas_stay_duration = str_to_title(overseas_stay_duration)
  ) |>
 select(c("overseas_stay_duration", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(overseas_stay_duration, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ overseas_stay_duration, scales = "free_y") +
 labs(
    x = "SA2 Rate of Overseas Stay Duration",
    y = "Victimization Rate",
    title = "Victimization Rate by Overseas Stay Duration"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.5 Education

6.5.1 Student Type

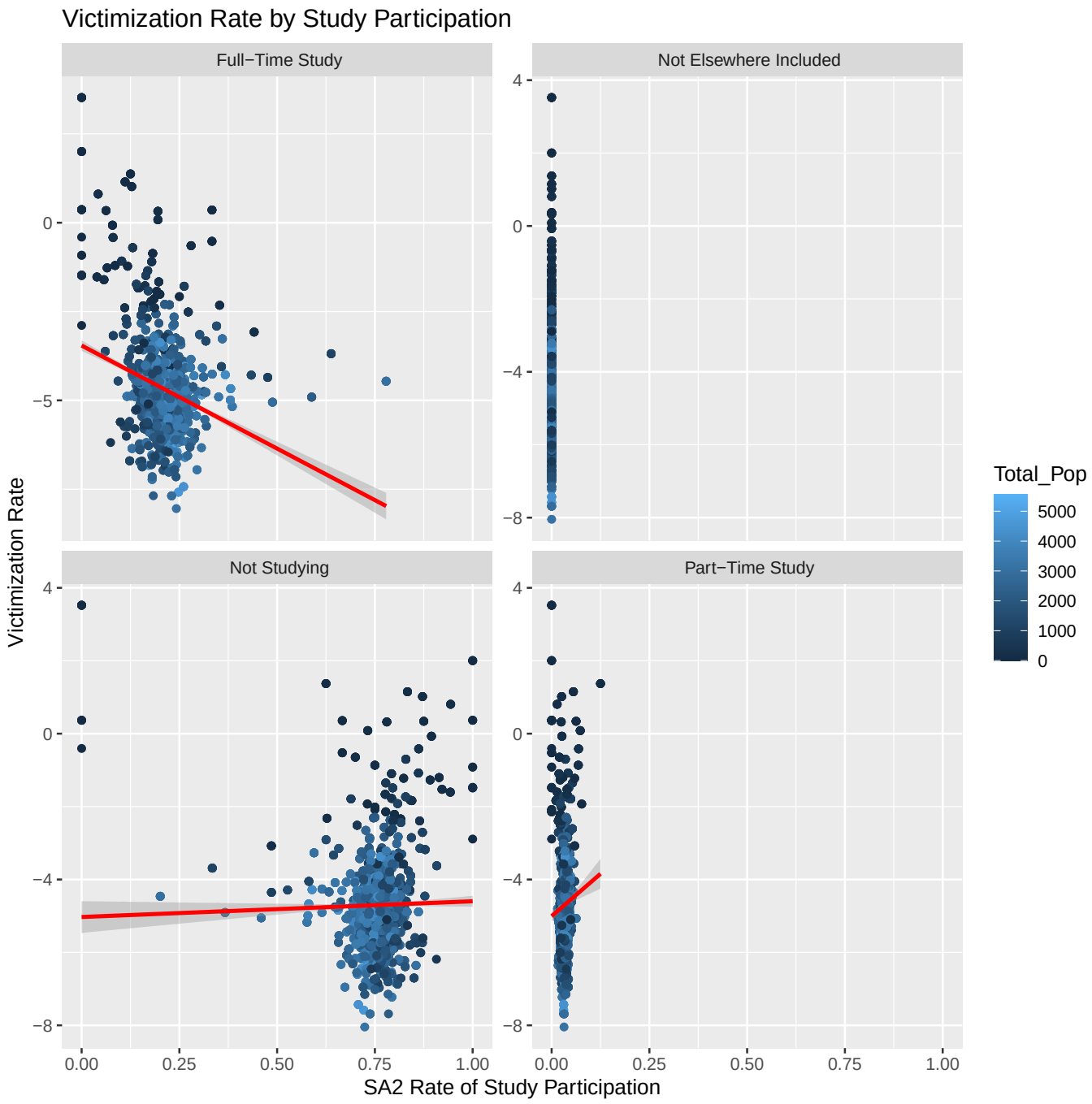
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Census usually resident population count, Year: 2023, Measure: Count, Var1
                "study_participation \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("study_participation"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
```

```
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
 pivot_longer(
    cols = starts_with("study_participation"),
    names_to = "study_participation",
    values_to = "proportion"
  ) |>
 mutate(
    study_participation = str_remove(study_participation, "study_participation"),
    study_participation = str_replace_all(study_participation, "_", " "),
    study_participation = str_to_title(study_participation)
  ) |>
 select(c("study_participation", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(study_participation, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ study_participation, scales = "free_y") +
 labs(
    x = "SA2 Rate of Study Participation",
    y = "Victimization Rate",
    title = "Victimization Rate by Study Participation"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.5.2 Education Qualifications

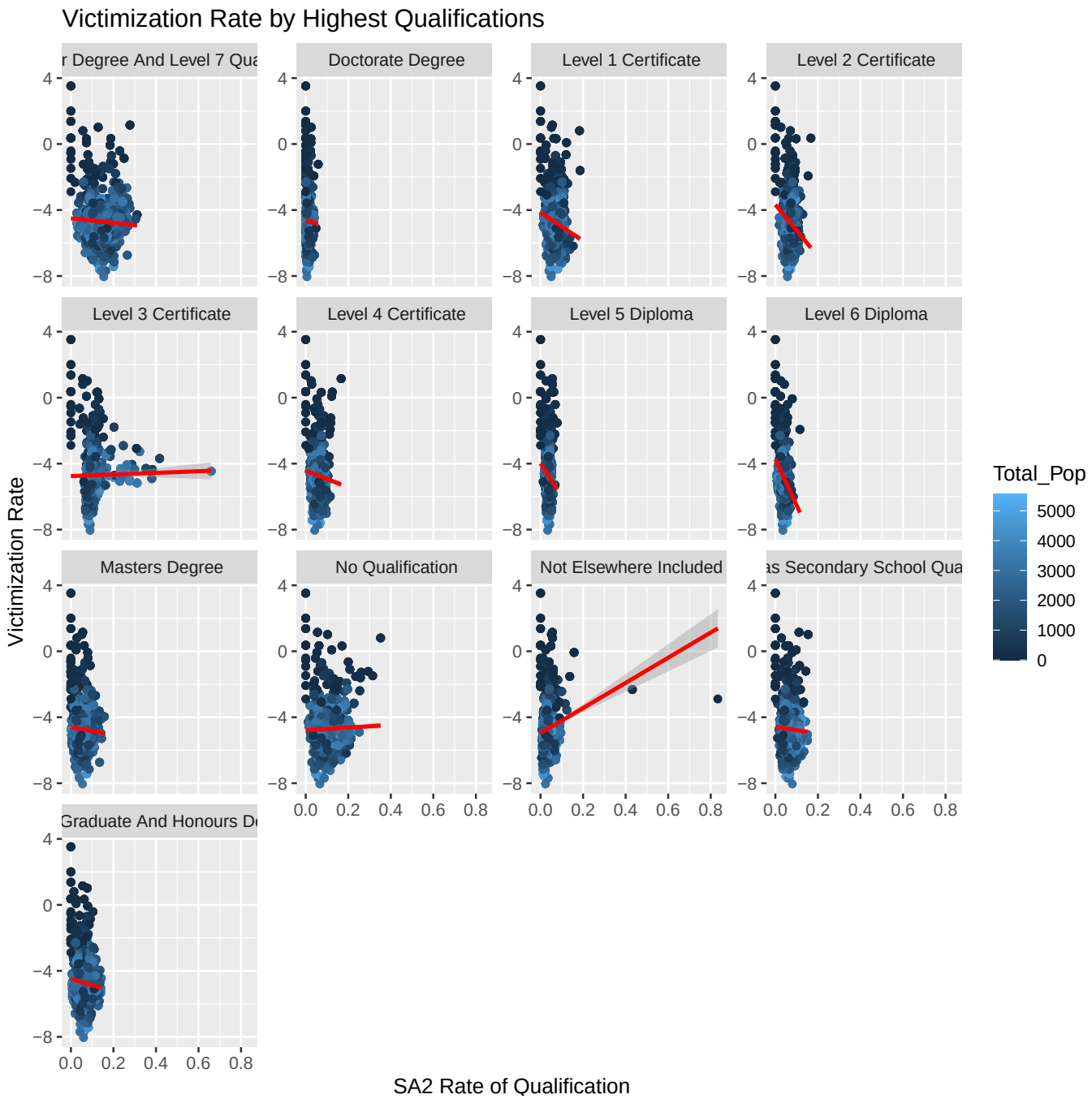
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Census usually resident population count aged 15 years and over, Year: 202
                "highest_qualification \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("highest_qualification"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
```

```
 pivot_longer(
    cols = starts_with("highest_qualification"),
    names_to = "highest_qualification",
    values_to = "proportion"
  ) |>
 mutate(
    highest_qualification = str_remove(highest_qualification, "highest_qualification"),
    highest_qualification = str_replace_all(highest_qualification, "_", " "),
    highest_qualification = str_to_title(highest_qualification)
  ) |>
 select(c("highest_qualification", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(highest_qualification, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ highest_qualification, scales = "free_y") +
 labs(
    x = "SA2 Rate of Qualification",
    y = "Victimization Rate",
    title = "Victimization Rate by Highest Qualifications"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
long |>
 mutate(
  highest_qualification = case_when(
 highest_qualification %in% c(" Masters Degree", " Post-Graduate And Honours Degrees", ' Doctorate Degree'
 highest_qualification %in% c(" Bachelor Degree And Level 7 Qualification") ~ "Bachelor",
 highest_qualification %in% c(" Level 6 Diploma", " Level 5 Diploma") ~ "Diploma",
 highest_qualification %in% c(" Level 4 Certificate", " Level 3 Certificate", " Level 2 Certificate", " Le
    TRUE ~ highest_qualification
  )
 ) |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ highest_qualification, scales = "free_y") +
```

```
  labs(
     x = "SA2 Rate of Qualification",
     y = "Victimization Rate",
     title = "Victimization Rate by Highest Qualifications"
   )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.6 Income

6.6.1 Source

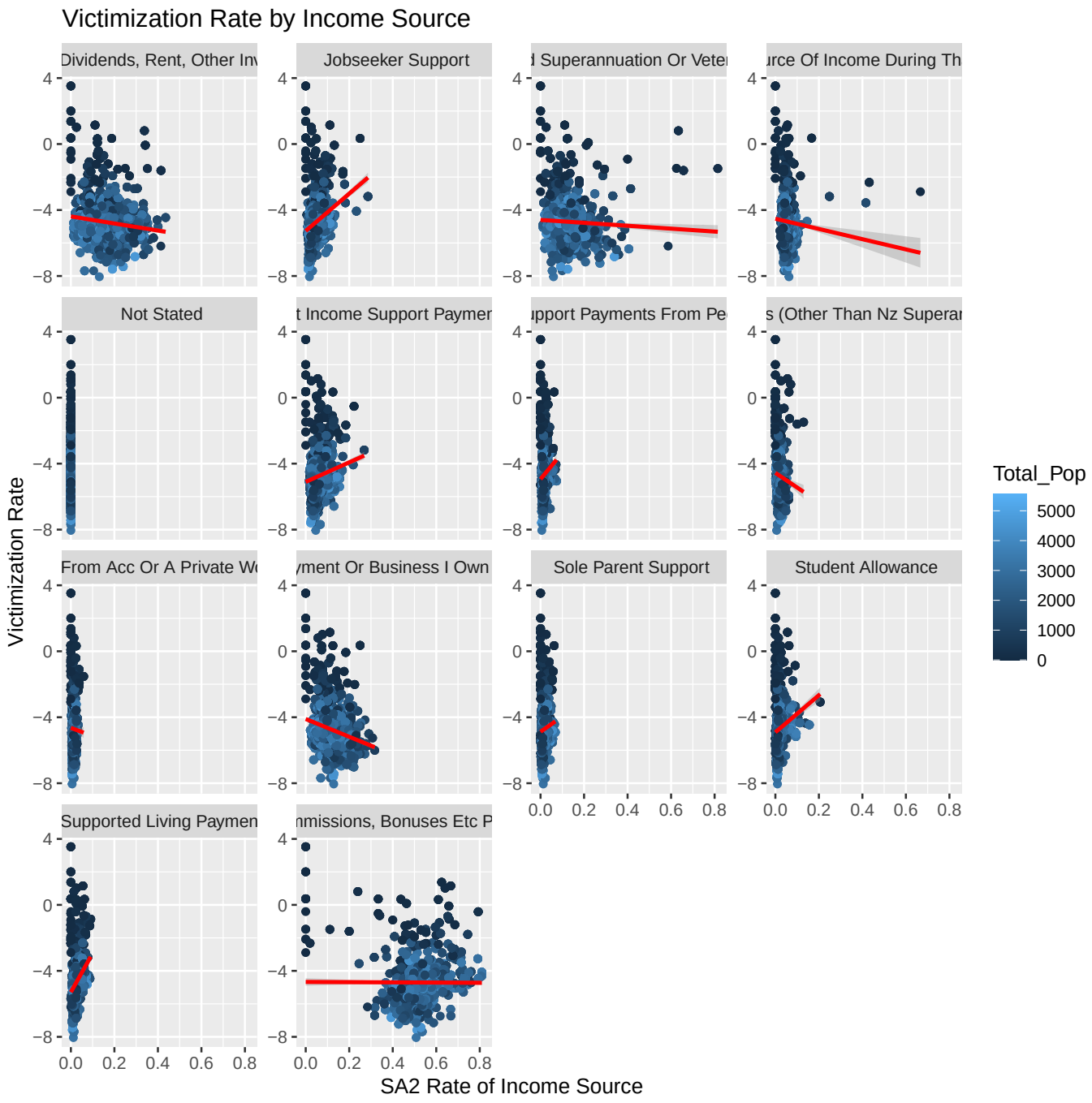
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
```

```
                  "Subject pop: Census usually resident population count aged 15 years and over, Year: 202
                  "income_source \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("income_source"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
 pivot_longer(
    cols = starts_with("income_source"),
    names_to = "income_source",
    values_to = "proportion"
  ) |>
 mutate(
    income_source = str_remove(income_source, "income_source"),
    income_source = str_replace_all(income_source, "_", " "),
    income_source = str_to_title(income_source)
  ) |>
 select(c("income_source", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(income_source, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ income_source, scales = "free_y") +
 labs(
    x = "SA2 Rate of Income Source",
    y = "Victimization Rate",
    title = "Victimization Rate by Income Source"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.6.2 Amount

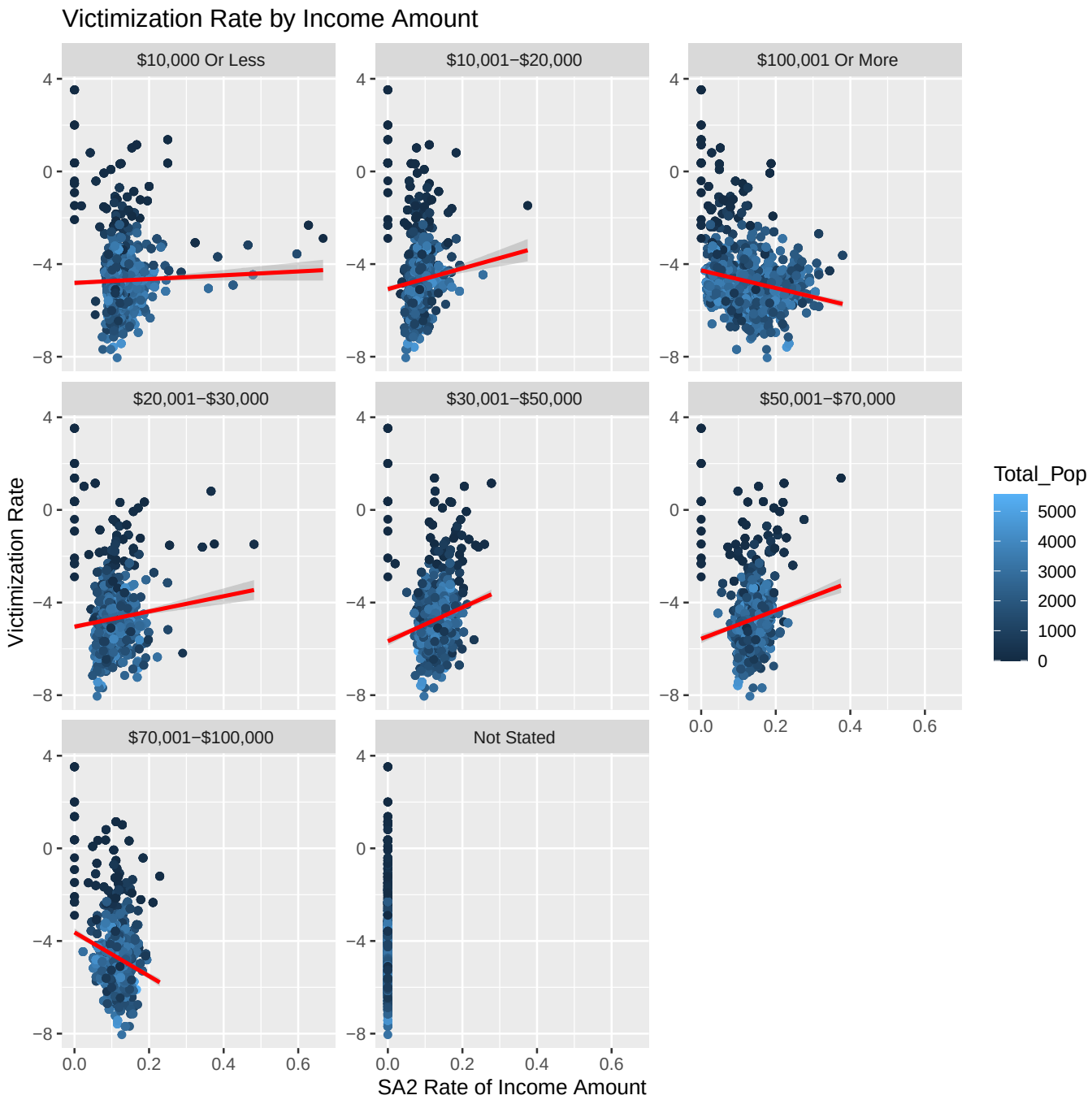
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Census usually resident population count aged 15 years and over, Year: 202
                "income_amount \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("income_amount"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
```

```
 pivot_longer(
    cols = starts_with("income_amount"),
    names_to = "income_amount",
    values_to = "proportion"
  ) |>
 mutate(
    income_amount = str_remove(income_amount, "income_amount"),
    income_amount = str_replace_all(income_amount, "_", " "),
    income_amount = str_to_title(income_amount)
  ) |>
 select(c("income_amount", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(income_amount, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ income_amount, scales = "free_y") +
 labs(
    x = "SA2 Rate of Income Amount",
    y = "Victimization Rate",
    title = "Victimization Rate by Income Amount"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.7 Job

6.7.1 Type

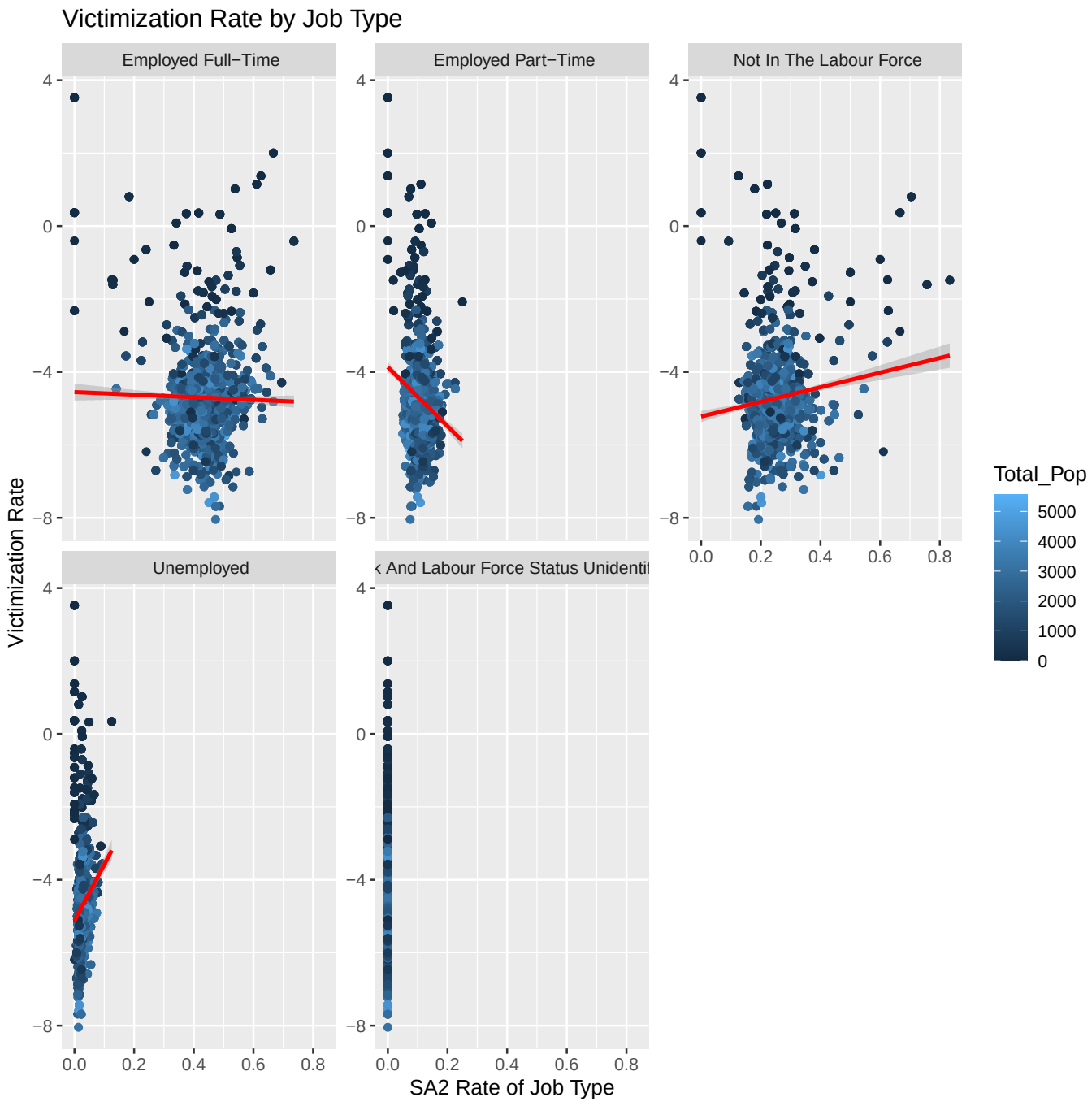
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Census usually resident population count aged 15 years and over, Year: 202
                "job_type \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("job_type"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
```

```
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
 pivot_longer(
    cols = starts_with("job_type"),
    names_to = "job_type",
    values_to = "proportion"
  ) |>
 mutate(
    job_type = str_remove(job_type, "job_type"),
    job_type = str_replace_all(job_type, "_", " "),
    job_type = str_to_title(job_type)
  ) |>
 select(c("job_type", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(job_type, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ job_type, scales = "free_y") +
 labs(
    x = "SA2 Rate of Job Type",
    y = "Victimization Rate",
    title = "Victimization Rate by Job Type"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.7.2 Industry

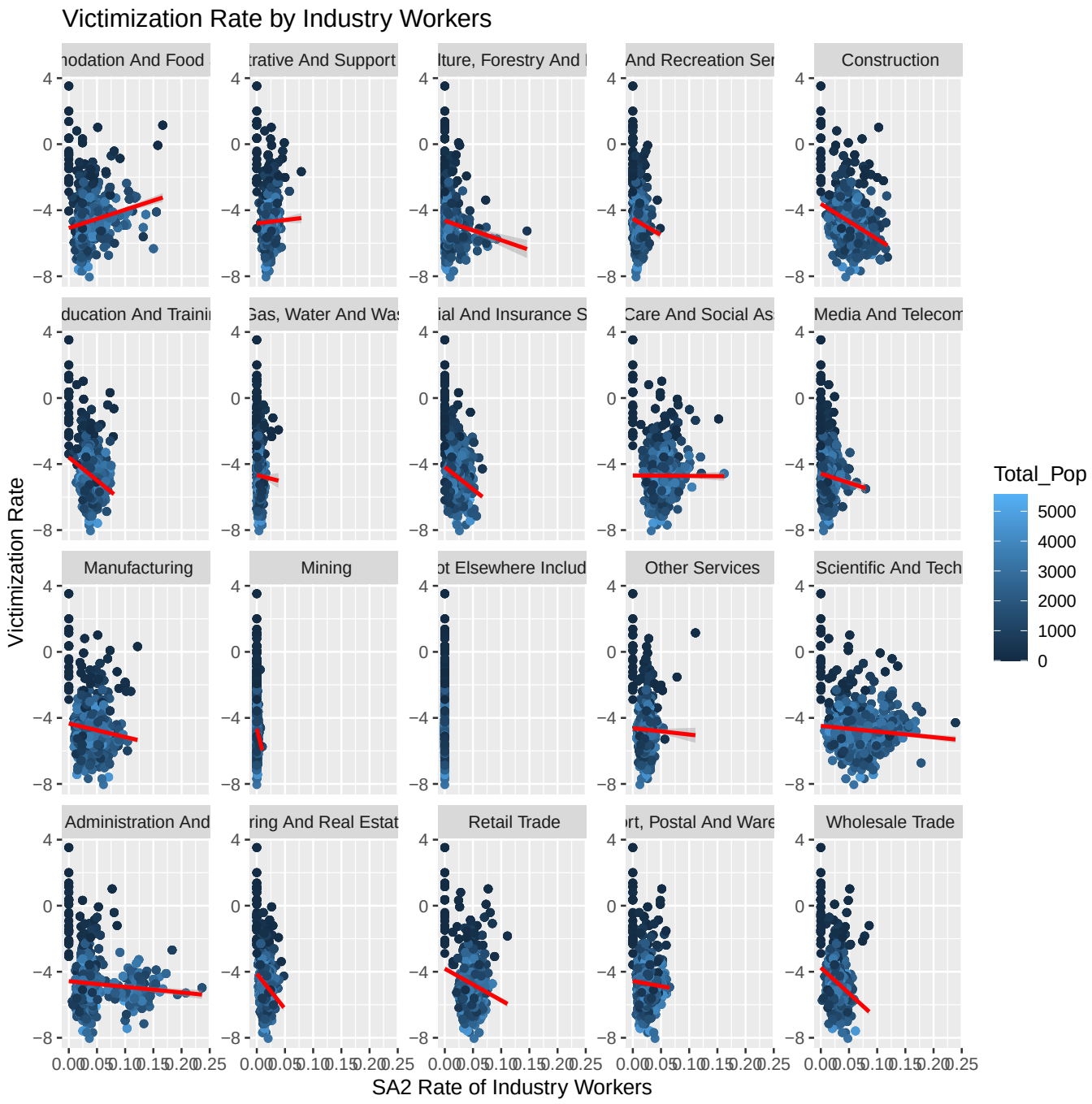
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Employed census usually resident population count aged 15 years and over,
                "industry \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("industry"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
```

```
 pivot_longer(
    cols = starts_with("industry"),
    names_to = "industry",
    values_to = "proportion"
  ) |>
 mutate(
    industry = str_remove(industry, "industry"),
    industry = str_replace_all(industry, "_", " "),
    industry = str_to_title(industry)
  ) |>
 select(c("industry", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(industry, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ industry, scales = "free_y") +
 labs(
    x = "SA2 Rate of Industry Workers",
    y = "Victimization Rate",
    title = "Victimization Rate by Industry Workers"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.7.3 Occupation

```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Employed census usually resident population count aged 15 years and over,
                "occupation \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("occupation"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
```

```
 pivot_longer(
    cols = starts_with("occupation"),
    names_to = "occupation",
    values_to = "proportion"
  ) |>
 mutate(
    occupation = str_remove(occupation, "occupation"),
    occupation = str_replace_all(occupation, "_", " "),
    occupation = str_to_title(occupation)
  ) |>
 select(c("occupation", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(occupation, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ occupation, scales = "free_y") +
 labs(
    x = "SA2 Rate of Occupation",
    y = "Victimization Rate",
    title = "Victimization Rate by Occupation"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



6.7.4 Sector

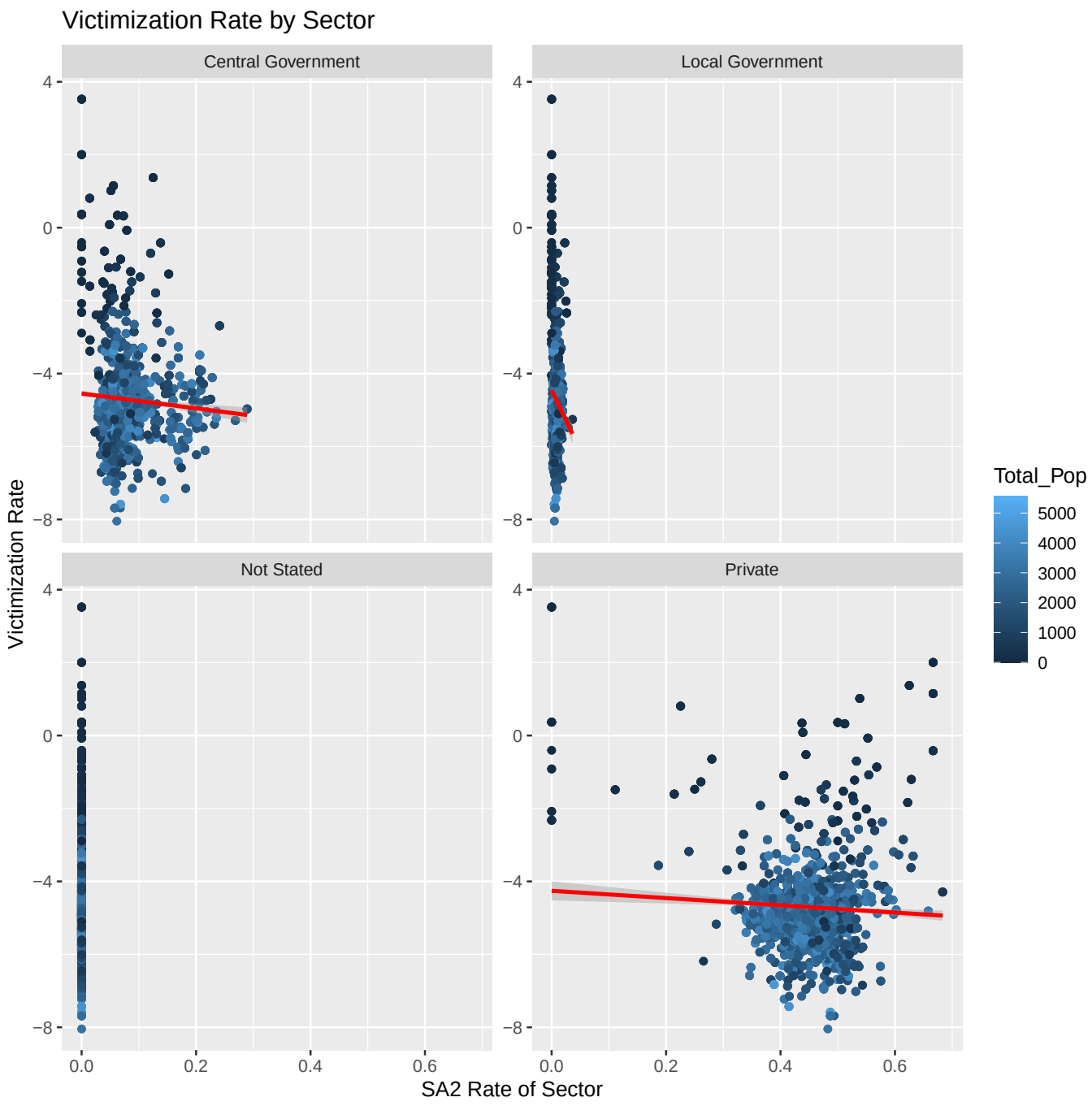
```
long = total_crime_sa2 |> rename_with(
  ~ str_replace(.x,
                "Subject pop: Employed census usually resident population count aged 15 years and over,
                "sector_ \\1") |>
 str_replace_all(" ", "_")
) |>
 group_by(SA2) |>
 mutate(
 across(starts_with("sector"), ~ sum(.x, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)),
    Vict_rate = sum(Victimisations, na.rm = TRUE) / sum(Total_Pop, na.rm = TRUE)
  ) |>
```

```
 pivot_longer(
    cols = starts_with("sector_"),
    names_to = "sector",
    values_to = "proportion"
  ) |>
 mutate(
    sector = str_remove(sector, "sector"),
    sector = str_replace_all(sector, "_", " "),
    sector = str_to_title(sector)
  ) |>
 select(c("sector", "proportion", "Vict_rate", "Total_Pop")) |>
  filter(!str_detect(sector, "Total"))
```

```
## Adding missing grouping variables: `SA2`
```

```
long |>
 ggplot(aes(x=proportion, y=log(Vict_rate), color = Total_Pop)) +
 geom_point() +
 geom_smooth(method = "lm", se = TRUE, color='red') +
 facet_wrap(~ sector, scales = "free_y") +
 labs(
    x = "SA2 Rate of Sector",
    y = "Victimization Rate",
    title = "Victimization Rate by Sector"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```


Victimization Rate by Sector
Central Government
Local Government
Not Stated
Private
Victimization Rate
SA2 Rate of Sector
Total_Pop
5000
4000
3000
2000
1000
0
4
0
−4
−8
0.0
0.2
0.4
0.6